

Cell Type Bacterial, gram positive

Species Used *Lactobacillus acidophilus*

Molecules Electroporated DNA: pACMI (or pACM2), pAMG10, pCK98, pBSKTAU, pGK12, pUB110. See Comments.

Before the Pulse

Cell Growth Medium Lactobacilli MRS broth (DIFCO Laboratories)

Growth Phase at Harvest O.D. (600) =1.0

Pre-pulse Incubation No incubation

Wash Solution See Comments

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 4 °C

Electroporation Medium pZB (1x)

Cuvette Gap 0.4 cm

Voltage 2.5 kV

Cell Density 2 x 10⁸ (8) to 8 x 10⁸ (8) cells / ml

Volume of Cells 0.8 ml

Field Strength 6.25 kV/cm

DNA Concentration 200 µg / ml

DNA Resuspension Buffer TE buffer (pH 8.0) (10mM Tris, 1mM EDTA)

Capacitor 25 µF

Volume of DNA 5 µl

Resistor 1000 Ω (Pulse Controller)

After the Pulse

Time Constant 3.0 to 4.8 msec

Outgrowth Medium Lactobacilli MRS broth (DIFCO)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
Wash solution: pZB(1X): 7 mM potassium phosphate (pH 7.4) containing 1 mM MgCl₂ and 272mM sucrose.

Selection Method or Assay Used: Antibiotic resistance method, pACMI (TcR), pAM610 (TcR) pCK98(KmR), pBSKTAU (KmR), pGK12(EmR), pUB110(KmR)

Ref: *Biotechnology Letters* (1990). Vol. 12. No. 12. 919-924.

Outgrowth Temperature 37 °C

Length of Incubation 3 hour

Selection Method or Assay Used See Comments

Electroporation Efficiency 3.1 x 10³ to 2.0 x 10³ (3) transformants / µg DNA (average)

Per Cent Survival 12%

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